Reflection Journal

1. Conceptual Breakthroughs: Deconstructing Complexity

The hands-on evaluation of polynomial degrees (1, 4, and 15) provided a clear visual illustration of the Goldilocks principle in predictive modeling. Initially, I assumed higher-degree models were inherently superior due to increased mathematical capacity. However, the degree 1 model demonstrated extreme simplicity by fitting a rigid line across a curved sine wave (high bias/underfitting). Conversely, the degree 15 model overfitted significantl, rather than extracting the structural trend, it aggressively chased individual noisy data points (high variance). The degree 4 polynomial struck the perfect balance, capturing the true wave pattern while remaining resilient to noise.

2. Diagnostic Tools: Decoding Learning Curves

Visualizing training vs cross-validation scores illuminated key diagnostic signals for model performance. At the first degree of underfitting, adding more data is useless for improving the score. For a high degree of overfitting (like degree 15) the scores also remain low with more data, but this time it is because the model is learning to memorize instead of generalize. This can be seen from the big gap between training and cross-validation scores.

3. Reflection on Strategy & Evolving Perspective

This lab changed my problem-solving approach. Rather than relying solely on raw training accuracy, I now evaluate the convergence gap between validation metrics to diagnose model health. Moving forward, I plan to explore regularization techniques such as Ridge and Lasso

regression to mathematically constrain feature coefficients, allowing complex models to generalize effectively without manual degree reduction.